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### MANUFACTURING METHOD OF POWER DEVICE HAVING DUAL POLYSILICON GATE

#### Abstract

A manufacturing method of a power device having a dual polysilicon gate, including: forming a well in a substrate; forming a gate oxide layer; forming a polysilicon gate layer; forming a photo resist layer on the polysilicon gate layer to define a reduced surface field region, an enhanced drift region, and a field plate groove; etching the polysilicon gate layer to form the field plate groove; implanting a plurality of first and second conductivity type dopants in the substrate to form the reduced surface field region and the enhanced drift region; forming a field plate region in the field plate groove; forming another polysilicon gate layer which connects and overlays the polysilicon gate layer and the field plate region; and etching the polysilicon gate layers to form a first polysilicon gate region and a second polysilicon gate region, so as to form the dual polysilicon gate.

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## Background/Summary

### CROSS REFERENCE

[0001] The present invention claims priority to TW 113108620 filed on Mar. 8, 2024.

### BACKGROUND OF THE INVENTION

#### Field of Invention

[0002] The present invention relates to a manufacturing method of a power device having dual polysilicon gate, particularly, it relates to such power device having dual polysilicon gate that can eliminate sharp angles at an interface between a gate and a field oxide region.

#### Description of Related Art

[0003] Referring to FIG. 1, which is a schematic cross-sectional view showing a prior art power device **10**. As shown in FIG. 1, the power device **10** is formed in a substrate **11** and comprises a gate oxide region **12**, a gate **13**, a field oxide region **17**, a spacer region **19**, a body region **20**, a source **21**, a drain **22**, and a shallow trench isolation (STI) region **23**.

[0004] The power device **10** is a type of lateral double-diffused metal-oxide-semiconductor (LDMOS) device. In typical LDMOS devices, the most effective way to improve the device breakdown voltage and mitigate the hot carrier injection problem near the gate edge is to adopt a field plate structure, which can alleviate the local electric field crowding that causes early device breakdown. Although the traditional LOCOS field oxide region (as the field oxide region **17** shown in FIG. 1) is the simplest way to realize the field plate structure without additional masking, the LOCOS field oxide region may lead to several severe issues, such as high ON resistance and diffusion of dopants (boron, phosphorus) in the LDMOS device into the field oxide region during high-temperature processes.

[0005] Furthermore, there exist sharp angles at the interface between the gate **13** and the gate oxide region **12** and the field oxide region **17**, which can cause process issues, such as voids or seams. Moreover, various oxide etching and re-deposition steps can aggravate the recession of the shallow trench isolation (STI) region **23**. The aforementioned dopants (boron, phosphorus) diffusion into the field oxide region **17** also affects the device characteristics and reliability.

[0006] In view of the above, the present invention proposes a manufacturing method of a power device having a dual polysilicon gate, which can avoid damage to the shallow trench isolation or LOCOS field oxide region, eliminate the sharp angles at the interface between the gate and the field oxide region, reduce the ON resistance, and improve the device characteristics and reliability.

### SUMMARY OF THE INVENTION

[0007] From one perspective, the present invention provides a manufacturing method of a power device having a dual polysilicon gate, comprising: forming a well in a substrate; forming a gate oxide layer directly connected and fully covering the substrate; forming a first polysilicon gate layer directly connected and fully covering the gate oxide layer; forming a first photoresist layer on the first polysilicon gate layer by a first photolithography process step to simultaneously define a reduced surface field region, an enhanced drift region, and a field plate groove; etching the first polysilicon gate layer by using the first photoresist layer as an etching mask to form the field plate groove; implanting a plurality of first conductivity type dopants into the substrate by using the first photoresist layer as an implanting mask to form the reduced surface field region; implanting a plurality of second conductivity type dopants into the substrate by using the first photoresist layer as an implanting mask to form the enhanced drift region, wherein the first and second conductivity type dopants have opposite electrical properties; forming a field plate region in the field plate

groove in a self-aligned process step; forming a second polysilicon gate layer directly connected to and fully covering the first polysilicon gate layer and the field plate region; and etching the first and second polysilicon gate layers to form a first polysilicon gate region and a second polysilicon gate region, so as to form the dual polysilicon gate; wherein a portion of the second polysilicon gate region is directly connected to and fully covers the first polysilicon gate region, and another portion of the second polysilicon gate region is directly connected to and covers a portion of the field plate region.

[0008] In one embodiment, the step of etching the first polysilicon gate layer to form the field plate groove by using the first photoresist layer as an etching mask is performed before the steps of implanting the plurality of first conductivity type dopants into the substrate by using the first photoresist layer as an implanting mask to form the reduced surface field region and implanting the plurality of second conductivity type dopants into the substrate by using the first photoresist layer as an implanting mask to form the enhanced drift region.

[0009] In one embodiment, the step of etching the first polysilicon gate layer to form the field plate groove by using the first photoresist layer as an etching mask is performed after the steps of implanting the plurality of first conductivity type dopants into the substrate by using the first photoresist layer as an implanting mask to form the reduced surface field region and implanting the plurality of second conductivity type dopants into the substrate by using the first photoresist layer as an implanting mask to form the enhanced drift region.

[0010] In one embodiment, the manufacturing method further comprises: forming a liner oxide layer by a deposition process step after forming the field plate groove, wherein the liner oxide layer fully covers the field plate groove.

[0011] In one embodiment, after forming the liner oxide layer, a liner oxide region and the field plate region are formed by a deposition process step and a chemical mechanical polishing process step, wherein the liner oxide region fully covers the field plate groove.

[0012] In one embodiment, after forming the liner oxide layer, an anisotropic etching process step is performed to etch the liner oxide layer to form a plurality of spacer oxide regions, wherein the plural spacer oxide regions cover a plurality of sidewalls of the field plate groove respectively but does not cover a bottom of the field plate groove.

[0013] In one embodiment, the manufacturing method further comprises: forming a body region in the substrate, wherein a portion of the body region is located directly below a portion of the first polysilicon gate region; and forming a source and a drain in the substrate below the dual polysilicon gate on both sides, wherein the source is located in the body region.

[0014] From another perspective, the present invention provides a manufacturing method of a power device having a dual polysilicon gate, comprising: forming a well in a substrate; forming a gate oxide layer directly connected and fully covering the substrate; forming a first polysilicon gate layer directly connected and fully covering the gate oxide layer; forming a first photoresist layer on the first polysilicon gate layer in a first photolithography process step to simultaneously define a reduced surface field region, an enhanced drift region, and a field plate groove; implanting a plurality of first conductivity type dopants into the substrate by using the first photoresist layer as an implanting mask to form the reduced surface field region; implanting a plurality of second conductivity type dopants into the substrate by using the first photoresist layer as an implanting mask to form the enhanced drift region, wherein the first and second conductivity type dopants have opposite electrical properties; implanting oxygen ions into the first polysilicon gate layer in an oxygen ion implantation process step to form the field plate region by using the first photoresist layer as an implanting mask; forming a second polysilicon gate layer directly connected to and fully covering the first polysilicon gate layer and the field plate region; and etching the first and second polysilicon gate layers to form a first polysilicon gate region and a second polysilicon gate region, so as to form the dual polysilicon gate; wherein a portion of the second polysilicon gate region is directly connected to and fully covers the first polysilicon gate region, and another portion

of the second polysilicon gate region is directly connected to and covers a portion of the field plate region.

[0015] In one embodiment, the manufacturing method further includes: forming a body region in the substrate, wherein a portion of the body region is located directly below a portion of the first polysilicon gate region; and forming a source and a drain in the substrate below the dual polysilicon gate on both sides, wherein the source is located in the body region.

[0016] The advantage of the present invention is that it can precisely control the profile and thickness of the field plate region, and since the field plate region does not require etching, there are no issues with oxide damage or purity.

[0017] The objectives, technical details, features, and effects of the present invention will be better understood with regard to the detailed description of the embodiments below, with reference to the attached drawings.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIG. 1 is a schematic cross-sectional view showing a prior art power device 10.

[0019] FIGS. 2A-2J are schematic cross-sectional views showing a manufacturing method of a power device 30 having a dual polysilicon gate according to an embodiment of the present invention.

[0020] FIGS. 3A-3H are schematic cross-sectional views showing a manufacturing method of a power device 50 having a dual polysilicon gate according to an embodiment of the present invention.

[0021] FIGS. 4A-4I are schematic cross-sectional views showing a manufacturing method of a power device 70 having a dual polysilicon gate according to an embodiment of the present invention.

[0022] FIGS. 5A-5G are schematic cross-sectional views showing a manufacturing method of a power device 90 having a dual polysilicon gate according to an embodiment of the present invention.

### DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0023] The drawings as referred to throughout the description of the present invention are for illustration only, to show the interrelations among the process steps and the layers, while the shapes, thicknesses, and widths are not drawn in actual scale.

[0024] Please refer to FIGS. 2A-2J, which show schematic cross-sectional views of a manufacturing method of a power device 30 having a dual polysilicon gate according to an embodiment of the present invention. The power device 30 having a dual polysilicon gate comprises: a substrate 31, a well 31a, a gate oxide region 32a, a first polysilicon gate region 33a, a reduced surface field region 34, an enhanced drift region 35, a liner oxide region 36a, a field plate region 37b, a second polysilicon gate region 38a, a gate spacer 39, a body region 40, a source 41, and a drain 42. As shown in FIG. 2A, a substrate 31 is provided first, and a well 31a is formed in the substrate 31. In one embodiment, a photoresist layer is formed on the substrate 31 in a photolithography process step to define the well 31a. Then, as shown in FIG. 2A, using the photoresist layer as a mask, for example, in an ion implantation process step, a plurality of impurities are implanted into the substrate 31 in the form of accelerated ions, as indicated by multiple arrows in FIG. 2A, to form the well 31a.

[0025] Next, as shown in FIG. 2B, a gate oxide layer 32 is formed directly connected to and fully covering the substrate 31. In one embodiment, the gate oxide layer 32 is formed by a thermal oxidation process step. Then, as shown in FIG. 2B, a first polysilicon gate layer 33 is formed directly connected to and fully covering the gate oxide layer 32. In one embodiment, the

polysilicon gate layer **33** is formed by a deposition process step. Subsequently, as shown in FIG. 2C, for example, in a first photolithography process step, a first photoresist layer **37'** is formed on the first polysilicon gate layer **33** to simultaneously define the reduced surface field region **34**, the enhanced drift region **35**, and the field plate groove **37a**. Next, as shown in FIG. 2C, using the first photoresist layer **37'** as a mask, for example, in an ion implantation process step, a plurality of first conductivity type dopants are implanted into the substrate **31** in the form of accelerated ions, as indicated by multiple arrows in FIG. 2C, to form the reduced surface field region **34**.

[0026] Then, as shown in FIG. 2C, using the first photoresist layer **37'** as a mask, for example, in an ion implantation process step, a plurality of second conductivity type dopants are implanted into the substrate **31** to form the enhanced drift region **35**. It should be noted that the first conductivity type dopants and the second conductivity type dopants have opposite electrical properties. Next, as shown in FIGS. 2C and 2D, using the first photoresist layer **37'** as a mask, the first polysilicon gate layer **33** is etched to form the field plate groove **37a**. Subsequently, as shown in FIGS. 2E and 2F, for example, in a self-aligned process step, the field plate region **37b** is formed in the field plate groove **37a**. In one embodiment, as shown in FIGS. 2D and 2E, after forming the field plate groove **37a**, for example, in a deposition process step, a liner oxide layer **36** is formed, wherein the liner oxide layer **36** fully covers the field plate groove **37a**. Then, as shown in FIGS. 2E and 2F, after forming the liner oxide layer **36**, for example, in a deposition process step and a chemical mechanical polishing process step, the liner oxide region **36a** and the field plate region **37b** are formed, wherein the liner oxide region **36a** fully covers the field plate groove **37a**. In one embodiment, the field plate region **37b** comprises a silicon oxide layer, a silicon nitride layer, a silicon oxynitride layer, and/or a high-k dielectric layer.

[0027] Next, as shown in FIG. 2G, a second polysilicon gate layer **38** is formed directly connected to and fully covering the first polysilicon gate layer **33** and the field plate region **37b**. Then, as shown in FIG. 2H, for example, in a photolithography process step and an anisotropic etching process step, the first polysilicon gate layer **33** and the second polysilicon gate layer **38** are etched to form a first polysilicon gate region **33a** and a second polysilicon gate region **38a**, thereby forming the dual polysilicon gate. As shown in FIG. 2H, a portion of the second polysilicon gate region **38a** is directly connected to and fully covers the first polysilicon gate region **33a**, and another portion of the second polysilicon gate region **38a** is directly connected to and covers a portion of the field plate region **37b**. Then, as shown in FIG. 2I, for example, in an ion implantation process step, a plurality of first conductivity type dopants are implanted into the substrate **31** to form the body region **40**, wherein a portion of the body region **40** is located directly below a portion of the first polysilicon gate region **33a**. Additionally, as shown in FIG. 2I, for example, in an ion implantation process step, a plurality of second conductivity type dopants are implanted into the substrate **31** to form an extension region **41a** of the source **41**. The extension region **41a** is a part of the source **41** and is located directly below the gate spacer **39** on the side near the source **41** formed subsequently.

[0028] Next, as shown in FIG. 2J, two gate spacers **39** is formed respectively on both sides of the dual polysilicon gate formed by the first polysilicon gate region **33a** and the second polysilicon gate region **38a**. Subsequently, as shown in FIG. 2J, the source **41** and the drain **42** are formed in the substrate **31** below both sides of the dual polysilicon gate, wherein the source **41** is located in the body region **40**.

[0029] The substrate **31** may be, but is not limited to, a P-type or N-type semiconductor substrate. This is well-known to those skilled in the art and is not described in detail here.

[0030] The well **31a** is of the first conductivity type. For example, in an ion implantation process step, N-type or P-type dopants are implanted into their respective defined regions in the form of accelerated ions. This is well-known to those skilled in the art and is not described in detail here. The first conductivity type may be P-type or N-type; when the first conductivity type is P-type, the second conductivity type is N-type; when the first conductivity type is N-type, the second

conductivity type is P-type.

[0031] In one embodiment, as shown in FIGS. 2C and 2D, the step of etching the first polysilicon gate layer **33** to form the field plate groove **37a** using the first photoresist layer **37'** as a mask is performed after the steps of implanting the plurality of first conductivity type dopants into the substrate **31** to form the reduced surface field region **34** and implanting the plurality of second conductivity type dopants into the substrate **31** to form the enhanced drift region **35** using the first photoresist layer **37'** as a mask.

[0032] FIGS. 3A-3H are schematic cross-sectional views showing a manufacturing method of a power device **50** having a dual polysilicon gate according to an embodiment of the present invention. The power device **50** having a dual polysilicon gate comprises: a substrate **51**, a well **51a**, a gate oxide region **52a**, a first polysilicon gate region **53a**, a reduced surface field region **54**, an enhanced drift region **55**, a plurality of spacer oxide regions **56a**, a field plate region **57b**, a second polysilicon gate region **58a**, a gate spacer **59**, a body region **60**, a source **61**, and a drain **62**. This embodiment is similar to the embodiment shown in FIGS. 2A-2J, with the difference that, as shown in FIGS. 3D and 3E, after forming the liner oxide layer **56**, an anisotropic etching process step is performed to etch the liner oxide layer **56** to form the plural spacer oxide regions **56a**, wherein the plural spacer oxide regions **56a** respectively cover plural the sidewalls of the field plate groove **57a** without covering the bottom of the field plate groove **57a**. Next, as shown in FIG. 3F, after forming the plural spacer oxide regions **56a**, a deposition process step and a chemical mechanical polishing process step are performed to form the field plate region **57b**. In this embodiment, for descriptions related to the gate oxide layer **52**, the first polysilicon gate layer **53**, the liner oxide layer **56**, the first photoresist layer **57'**, the field plate groove **57a**, the second polysilicon gate layer **58**, and the extension region **61a**, please refer to the descriptions of the gate oxide layer **32**, the first polysilicon gate layer **33**, the liner oxide layer **36**, the first photoresist layer **37'**, the field plate groove **37a**, the second polysilicon gate layer **38**, and the extension region **41a** in the embodiment shown in FIGS. 2A-2J.

[0033] FIGS. 4A-4I are schematic cross-sectional views showing a manufacturing method of a power device **70** having a dual polysilicon gate according to an embodiment of the present invention. The power device **70** having a dual polysilicon gate comprises: a substrate **71**, a well **71a**, a gate oxide region **72a**, a first polysilicon gate region **73a**, a reduced surface field region **74**, an enhanced drift region **75**, a plurality of spacer oxide regions **76a**, a field plate region **77b**, a second polysilicon gate region **78a**, a gate spacer **79**, a body region **80**, a source **81**, and a drain **82**. This embodiment is similar to the embodiment shown in FIGS. 3A-3H, with the difference that, as shown in FIGS. 4C and 4D, the step of etching the first polysilicon gate layer **73** to form the field plate groove **77a** using the first photoresist layer **77'** as a mask is performed before the steps of implanting the plurality of first conductivity type dopants into the substrate **71** to form the reduced surface field region **74** and implanting the plurality of second conductivity type dopants into the substrate **71** to form the enhanced drift region **75** using the first photoresist layer **77'** as a mask. In this embodiment, for descriptions related to the gate oxide layer **72**, the first polysilicon gate layer **73**, the liner oxide layer **76**, the first photoresist layer **77'**, the field plate groove **77a**, the second polysilicon gate layer **78**, and the extension region **81a**, please refer to the descriptions of the gate oxide layer **32**, the first polysilicon gate layer **33**, the liner oxide layer **36**, the first photoresist layer **37'**, the field plate groove **37a**, the second polysilicon gate layer **38**, and the extension region **41a** in the embodiment shown in FIGS. 2A-2J.

[0034] FIGS. 5A-5G are schematic cross-sectional views showing a manufacturing method of a power device **90** having a dual polysilicon gate according to an embodiment of the present invention. The power device **90** having a dual polysilicon gate comprises: a substrate **91**, a well **91a**, a gate oxide region **92a**, a first polysilicon gate region **93a**, a reduced surface field region **94**, an enhanced drift region **95**, a field plate region **97b**, a second polysilicon gate region **98a**, a gate spacer **99**, a body region **100**, a source **101**, and a drain **102**. This embodiment is similar to the

embodiment shown in FIGS. 2A-2I, with the difference that, as shown in FIG. 5D, the oxygen ions are implanted into the first polysilicon gate layer **93** using the first photoresist layer **97'** as a mask, for example, in an oxygen ion implantation process step to form the field plate region **97b**, and this embodiment does not require forming a liner oxide region or spacer oxide regions. In this embodiment, for descriptions related to the gate oxide layer **92**, the first polysilicon gate layer **93**, the first photoresist layer **97'**, the field plate groove **97a**, the second polysilicon gate layer **98**, and the extension region **101a**, please refer to the descriptions of the gate oxide layer **32**, the first polysilicon gate layer **33**, the liner oxide layer **36**, the first photoresist layer **37'**, the field plate groove **37a**, the second polysilicon gate layer **38**, and the extension region **41a** in the embodiment shown in FIGS. 2A-2J.

[0035] In summary, the present invention can precisely control the profile and thickness of the field plate region, and since the field plate region does not require etching, there are no issues with oxide damage or purity.

[0036] The present invention has been described in considerable detail with reference to certain preferred embodiments thereof. It should be understood that the description is for illustrative purpose, not for limiting the broadest scope of the present invention. Those skilled in this art can readily conceive variations and modifications within the spirit of the present invention. The various embodiments described above are not limited to being used alone; two embodiments may be used in combination, or a part of one embodiment may be used in another embodiment. For example, other process steps or structures, such as a deep well region, may be added. For another example, the lithography process step is not limited to the mask technology but it can also include electron beam lithography, immersion lithography, etc. Therefore, in the same spirit of the present invention, those skilled in the art can t various equivalent variations and various combinations, and there are many combinations thereof, and the description will not be repeated here. The scope of the present invention should include what are defined in the claims and the equivalents.

## Claims

**1.** A manufacturing method of a power device having a dual polysilicon gate, comprising: forming a well in a substrate; forming a gate oxide layer directly connected and fully covering the substrate; forming a first polysilicon gate layer directly connected and fully covering the gate oxide layer; forming a first photoresist layer on the first polysilicon gate layer by a first photolithography process step to simultaneously define a reduced surface field region, an enhanced drift region, and a field plate groove; etching the first polysilicon gate layer by using the first photoresist layer as an etching mask to form the field plate groove; implanting a plurality of first conductivity type dopants into the substrate by using the first photoresist layer as an implanting mask to form the reduced surface field region; implanting a plurality of second conductivity type dopants into the substrate by using the first photoresist layer as an implanting mask to form the enhanced drift region, wherein the first and second conductivity type dopants have opposite electrical properties; forming a field plate region in the field plate groove in a self-aligned process step; forming a second polysilicon gate layer directly connected to and fully covering the first polysilicon gate layer and the field plate region; and etching the first and second polysilicon gate layers to form a first polysilicon gate region and a second polysilicon gate region, so as to form the dual polysilicon gate; wherein a portion of the second polysilicon gate region is directly connected to and fully covers the first polysilicon gate region, and another portion of the second polysilicon gate region is directly connected to and covers a portion of the field plate region.

**2.** The manufacturing method of claim 1, wherein the step of etching the first polysilicon gate layer to form the field plate groove by using the first photoresist layer as an etching mask is performed before the steps of implanting the plurality of first conductivity type dopants into the substrate by using the first photoresist layer as an implanting mask to form the reduced surface field region and

implanting the plurality of second conductivity type dopants into the substrate by using the first photoresist layer as an implanting mask to form the enhanced drift region.

**3.** The manufacturing method of claim 1, wherein the step of etching the first polysilicon gate layer to form the field plate groove by using the first photoresist layer as an etching mask is performed after the steps of implanting the plurality of first conductivity type dopants into the substrate by using the first photoresist layer as an implanting mask to form the reduced surface field region and implanting the plurality of second conductivity type dopants into the substrate by using the first photoresist layer as an implanting mask to form the enhanced drift region.

**4.** The manufacturing method of claim 1, further comprising: forming a liner oxide layer by a deposition process step after forming the field plate groove, wherein the liner oxide layer fully covers the field plate groove.

**5.** The manufacturing method of claim 4, wherein after forming the liner oxide layer, a liner oxide region and the field plate region are formed by a deposition process step and a chemical mechanical polishing process step, wherein the liner oxide region fully covers the field plate groove.

**6.** The manufacturing method of claim 4, wherein after forming the liner oxide layer, an anisotropic etching process step is performed to etch the liner oxide layer to form a plurality of spacer oxide regions, wherein the plural spacer oxide regions cover a plurality of sidewalls of the field plate groove respectively but does not cover a bottom of the field plate groove.

**7.** The manufacturing method of claim 1, wherein the field plate region comprises a silicon oxide layer, a silicon nitride layer, a silicon oxynitride layer, and/or a high-k dielectric layer.

**8.** The manufacturing method of claim 1, further comprising: forming a body region in the substrate, wherein a portion of the body region is located directly below a portion of the first polysilicon gate region; and forming a source and a drain in the substrate below the dual polysilicon gate on both sides, wherein the source is located in the body region.

**9.** A manufacturing method of a power device having a dual polysilicon gate, comprising: forming a well in a substrate; forming a gate oxide layer directly connected and fully covering the substrate; forming a first polysilicon gate layer directly connected and fully covering the gate oxide layer; forming a first photoresist layer on the first polysilicon gate layer in a first photolithography process step to simultaneously define a reduced surface field region, an enhanced drift region, and a field plate groove; implanting a plurality of first conductivity type dopants into the substrate by using the first photoresist layer as an implanting mask to form the reduced surface field region; implanting a plurality of second conductivity type dopants into the substrate by using the first photoresist layer as an implanting mask to form the enhanced drift region, wherein the first and second conductivity type dopants have opposite electrical properties; implanting oxygen ions into the first polysilicon gate layer in an oxygen ion implantation process step to form the field plate region by using the first photoresist layer as an implanting mask; forming a second polysilicon gate layer directly connected to and fully covering the first polysilicon gate layer and the field plate region; and etching the first and second polysilicon gate layers to form a first polysilicon gate region and a second polysilicon gate region, so as to form the dual polysilicon gate; wherein a portion of the second polysilicon gate region is directly connected to and fully covers the first polysilicon gate region, and another portion of the second polysilicon gate region is directly connected to and covers a portion of the field plate region.

**10.** The manufacturing method of claim 9, further comprising: forming a body region in the substrate, wherein a portion of the body region is located directly below a portion of the first polysilicon gate region; and forming a source and a drain in the substrate below the dual polysilicon gate on both sides, wherein the source is located in the body region.

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